Visualización de Series de Tiempo

Gestión de Datos en Entornos Inteligentes (XDEI)

Time Series Data

Visualización de Series de Tiempo

Time Series Data

- **QuantumLeap**
  - Componente FIWARE que almacena los **datos históricos** del CB
  - Permite consulta avanzada
  - Utiliza CrateDB o TimescaleDB como respaldo
- **Grafana**
  - Herramienta de **visualización avanzada** de **fuentes de datos diversas**
  - Nos permite visualizar los datos históricos almacenados por QuantumLeap en CrateDB

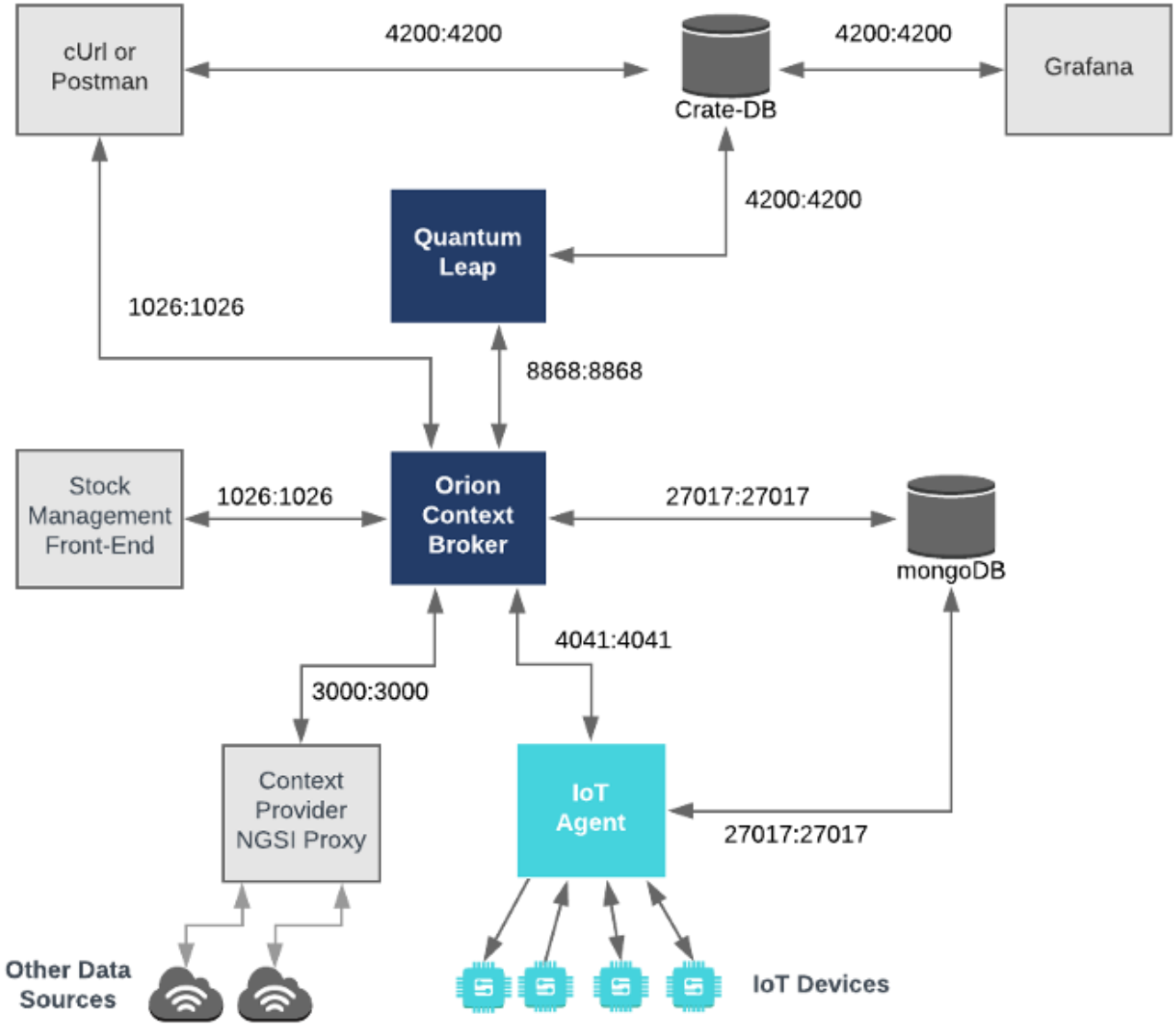
- Tutorial

```
git clone https://github.com/FIWARE/tutorials.Time-Series-Data.git
cd tutorials.Time-Series-Data
```

  - Copiar ficheros del campus virtual

```
./services start
```

Time Series Data
Arquitectura del tutorial



Time Series Data
Suscripciones de QL al CB

- **QL se suscribe al CB para recibir las actualizaciones** de los datos de contexto y almacenarlas como histórico
- Suscripción a los eventos del sensor de movimiento

```
curl -iX POST 'http://localhost:1026/v2/subscriptions/' \
  -H 'Content-Type: application/json' \
  -H 'fiware-service: openiot' -H 'fiware-servicepath: /' \
  -d '{ "description":
           "Notify QuantumLeap of count changes of any Motion Sensor",
        "subject": {
            "entities": [ { "idPattern": "Motion.*" } ],
            "condition": { "attrs": [ "count" ] }
        },
        "notification": {
           "http": { "url": "http://quantumleap:8668/v2/notify" },
           "attrs": [ "count" ],
           "metadata": ["dateCreated", "dateModified"]
        }
}'
```

Time Series Data
Suscripciones de QL al CB (II)

- Suscripción a los valores de luminosidad de la lámpara
  - *throttling* indica cada cuántos segundos enviar la notificación

```
curl -iX POST 'http://localhost:1026/v2/subscriptions/' \
  -H 'Content-Type: application/json' \
  -H 'fiware-service: openiot' -H 'fiware-servicepath: /' \
  -d '{ "description":
           "Notify QuantumLeap on luminosity changes on any Lamp",
  "subject": {
    "entities": [ { "idPattern": "Lamp.*" } ],
    "condition": { "attrs": [ "luminosity", "location" ] }
  },
  "notification": {
    "http": { "url": "http://quantumleap:8668/v2/notify" },
    "attrs": [ "luminosity", "location" ],
    "metadata": ["dateCreated", "dateModified"]
  },
  "throttling": 1
}'
```

Time Series Data
Suscripciones de QL al CB (III)

- El *throttling* puede **eliminar notificaciones pendientes** del CB a QL
  - En el caso de la lámpara, se puede comprobar que acaban llegando más notificaciones a QL según la lámpara, en este orden: Lamp:001 > Lamp:002 > Lamp:003 > Lamp:004
  - Esto hace necesario crear **una suscripción independiente para cada lámpara**
- Para que se produzcan actualizaciones en el sensor de movimiento y la luminosidad de la lámpara y lleguen a QL, es necesario **abrir la puerta** (para desbloquearla) y **encender la lámpara**

- Listado de suscripciones

```
curl -X GET 'http://localhost:1026/v2/subscriptions/' \
  -H 'fiware-service: openiot' -H 'fiware-servicepath: /'
```

Time Series Data
Consultar datos históricos (QL API)

- QL ofrece el API estándar NGSI de consulta
- **Primeros *N* valores**

```
curl -X GET \
  'http://localhost:8668/v2/entities/Lamp:001/attrs/luminosity?limit=3' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot' -H 'Fiware-ServicePath: /'
```

- Resp.:


```
{
    "attrName": "luminosity",
    "entityId": "Lamp:001",
    "entityType": "Lamp",
    "index": ["2023-07-18T16:21:37.915+00:00", "2023-07-
18T16:21:42.800+00:00", "2023-07-18T16:21:47.851+00:00"],
    "values": [1992, 1996, 1995]
}
```

Time Series Data
Consultar datos históricos (QL API) (II)

- **Primeros *N* valores desde un *offset***

```
curl -X GET \
'http://localhost:8668/v2/entities/Motion:001/attrs/count?offset=3&limit=3' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot' -H 'Fiware-ServicePath: /'
```

- **Últimos *N* valores**

```
curl -X GET \
  'http://localhost:8668/v2/entities/Motion:001/attrs/count?lastN=3' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot'  -H 'Fiware-ServicePath: /'
```

Time Series Data
Consultar datos históricos (QL API) (III)

- **Sumas de valores agrupados por período temporal**
  - Req. Versión >= 0.4.1. Comprobación de versión mediante

```
curl -X GET 'http://localhost:8668/version’ \
  -H 'Accept: application/json'
```

  - Petición

```
curl -X GET
'http://localhost:8668/v2/entities/Motion:001/attrs/count
?aggrMethod=count&aggrPeriod=minute&lastN=3' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot' -H 'Fiware-ServicePath: /'
```

  - Resp.:

```
{
    "attrName": "count",
    "entityId": "Motion:001", "entityType": "Motion",
    "index": ["2023-07-18T16:23:00.000+00:00",
              "2023-07-18T16:24:00.000+00:00",
              "2023-07-18T16:25:00.000+00:00"],
    "values": [20, 14, 21]
}
```

Time Series Data
Consultar datos históricos (QL API) (IV)

- **Valor máximo en un período temporal**

```
curl -X GET
'http://localhost:8668/v2/entities/Lamp:001/attrs/luminosity?aggrMethod
=max&fromDate=<previous_time>&toDate=<current_time>' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot' -H 'Fiware-ServicePath: /'
```

  - *<previous_time>* y *<current_time>* hay que sustituirlos por una fecha en formato ISO 8601, p. ej. 2024-04-04T17:34:26.000Z
  - Resp.:

```
{
    "attrName": "luminosity",
    "entityId": "Lamp:001",
    "entityType": "Lamp",
    "index": [],
    "values": [1999]
}
```

Time Series Data
Consultar datos históricos (QL API) (V)

- **Últimos N valores de los dispositivos cerca de un punto**

```
curl -X GET
'http://localhost:8668/v2/types/Lamp/attrs/luminosity?lastN=4&georel=
near;maxDistance:5000&geometry=point&coords=52.5547,13.3986' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot' -H 'Fiware-ServicePath: /'
```

- Resp.:


```
{ "data": {
      "attrName": "luminosity",
      "entities": [
          { "entityId": "Lamp:001",
            "index": [ "2018-12-13T16:35:58.284",
                       "2018-12-13T16:36:58.216" ],
            "values": [999, 999] },
          { "entityId": "Lamp:004",
            "index": [ "2018-12-13T16:35:04.351",
                       "2018-12-13T16:36:04.282" ],
            "values": [948, 948] } ],
       "entityType": "Lamp" } }
```

Time Series Data
Consultar datos históricos (QL API) (VI)

- **Últimos N valores de los dispositivos en un área geográfica**

```
curl -X GET
'http://localhost:8668/v2/types/Lamp/attrs/luminosity?lastN=4&georel=
coveredBy&geometry=polygon&coords=52.5537,13.3996;52.5557,13.3996;52.
5557,13.3976;52.5537,13.3976;52.5537,13.3996' \
  -H 'Accept: application/json' \
  -H 'Fiware-Service: openiot'  -H 'Fiware-ServicePath: /'
```

- Resp.:


```
{ "data": {
            "attrName": "luminosity",
            "entities": [ {
                 "entityId": "Lamp:001",
                 "index": [ "2018-12-13T17:08:56.041",
                            "2018-12-13T17:09:55.976",
                            "2018-12-13T17:10:55.907",
                            "2018-12-13T17:11:55.833" ],
                      "values": [999, 999, 999, 999]
                   } ],
            "entityType": "Lamp" } }
```

Time Series Data
Consultar datos históricos (SQL CrateDB)

- Podemos usar **SQL para consultar CrateDB**
  - Desde un lenguaje de programación usando un *driver*
  - Mediante HTTP contra http://<cratedb-host>:4200/_sql
  - En la interfaz web de CrateDB en http://<cratedb-host>:4200

Time Series Data
Consultar histórico (SQL CrateDB) (II)

CrateDB | Nodes: 1 | Health: Data Checks | System load: 0.43/ 0.44/ 0.20 | crate

Console

Enter a SQL statement to query CrateDB | Format results | Store console history persistently | Show error trace | Clear History

```
SELECT entity_id, entity_type, time_index, fiware_servicepath, luminosity, location
FROM "mtopeniot"."etlamp"
ORDER BY time_index DESC
LIMIT 100;
```

SUBMIT QUERY | Hint: Press ⇧ + ⏎ to submit query | SELECT OK, 20 records returned (0.035 seconds)

Result from query | 1 / 1

entity_id	entity_type	time_index	fiware_servicepath	luminosity	location
Lamp:001	Lamp	1712341562101 (2024-04-05T18:26:02.101Z)	/	1996	Type: Point Coordinates: [13.3986,52.5547]
Lamp:001	Lamp	1712341557106 (2024-04-05T18:25:57.106Z)	/	1991	Type: Point Coordinates: [13.3986,52.5547]
Lamp:001	Lamp	1712341552114 (2024-04-05T18:25:52.114Z)	/	1994	Type: Point Coordinates: [13.3986,52.5547]

Time Series Data
Consulta histórico (SQL CrateDB) (III)

- **Comprobación de la persistencia**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SHOW SCHEMAS"}'
```

- Resp.:

```
{   "cols": ["schema_name"],
    "rows": [["blob"], ["doc"], ["information_schema"],
             ["mtopeniot"], ["pg_catalog"], ["sys"]],
    "rowcount": 6,
    "duration": 15.559625}
```

- **Comprobación de la persistencia (II)**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SHOW TABLES"}'
```

- Resp.:

```
{   "cols": ["table_name"],
    "rows": [["etlamp"], ["etmotion"], ["md_ets_metadata"]],
    "rowcount": 3,
    "duration": 15.263458 }
```

Time Series Data
Consulta histórico (SQL CrateDB) (IV)

- ***N* primeros valores**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SELECT * FROM mtopeniot.etlamp WHERE entity_id =
'\''Lamp:001'\'' ORDER BY time_index ASC LIMIT 3"}'
```

- Resp.:


```
{ "cols": ["entity_id", "entity_type", "time_index",
           "fiware_servicepath", "__original_ngsi_entity__",
           "instanceid", "count"],
  "rows": [["Motion:001", "Motion", 1689697299869, "/", null,
            "urn:ngsi-ld:6232dc49-8a03-4471-8d49-266d2f5b4234", 0],
           ["Motion:001", "Motion", 1689697300930, "/", null,
            "urn:ngsi-ld:65d82c77-01c4-483b-bda1-7496feb7bd0f", 1],
           ["Motion:001", "Motion", 1689697301857, "/", null,
            "urn:ngsi-ld:90f2db3d-1cb0-48e4-ab8c-c71e4ceac1d3", 0]],
  "rowcount": 3, "duration": 301.9434}
```

Time Series Data
Consulta histórico (SQL CrateDB) (V)

- ***N* últimos valores**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SELECT * FROM mtopeniot.etmotion WHERE entity_id =
'\''Motion:001'\''  ORDER BY time_index DESC LIMIT 3"}'
```

- Resp.:


```
{ "cols": ["entity_id", "entity_type", "time_index",
           "fiware_servicepath", "__original_ngsi_entity__",
           "instanceid", "count"],
  "rows": [ ["Motion:001", "Motion", 1689699097452, "/", null,
             "urn:ngsi-ld:99910d16-b388-4216-ada9-4c2df8feef8a", 0],
            ["Motion:001", "Motion", 1689699095481, "/", null,
             "urn:ngsi-ld:070ef58d-36b5-46b1-a9ef-72897972718e", 1],
            ["Motion:001", "Motion", 1689699093507, "/", null,
             "urn:ngsi-ld:7bf0f277-0203-4912-ba5b-2555bafd661b", 0]],
  "rowcount": 3, "duration": 783.8505}
```

Time Series Data
Consulta histórico (SQL CrateDB) (VI)

- **Suma de valores agrupados por período temporal**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SELECT DATE_FORMAT (DATE_TRUNC ('\''minute'\'',
time_index)) AS minute, SUM (count) AS sum FROM mtopeniot.etmotion
WHERE entity_id = '\''Motion:001'\'' GROUP BY minute LIMIT 3"}'
```

- Resp.:


```
{
    "cols": ["minute", "sum"],
    "rows": [
        ["2023-07-18T16:26:00.000000Z", 11],
        ["2023-07-18T16:36:00.000000Z", 11],
        ["2023-07-18T16:29:00.000000Z", 14]
    ],
    "rowcount": 3,
    "duration": 867.4263
}
```

Time Series Data
Consulta histórico (SQL CrateDB) (VII)

- **Mínimo de valores agrupados por período temporal**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SELECT DATE_FORMAT (DATE_TRUNC ('\''minute'\'',
time_index)) AS minute, MIN (luminosity) AS min FROM mtopeniot.etlamp
WHERE entity_id = '\''Lamp:001'\'' GROUP BY minute"}'
```

- Resp.:


```
{
    "cols":["minute","min"],
    "rows":[
        ["2023-07-18T16:26:00.000000Z",1834],
        ["2023-07-18T16:36:00.000000Z",1722],
        ["2023-07-18T16:29:00.000000Z",1881],
        ["2023-07-18T16:32:00.000000Z",1866],
        ["2023-07-18T16:23:00.000000Z",1684],
        ... etc
    ],
    "rowcount":33, "duration":46.888416
}
```

Time Series Data
Consulta histórico (SQL CrateDB) (VIII)

- **Máximo valor de un período temporal**

```
curl -iX POST 'http://localhost:4200/_sql' \
  -H 'Content-Type: application/json' \
  -d '{"stmt":"SELECT MAX(luminosity) AS max FROM mtopeniot.etlamp
WHERE entity_id = '\''Lamp:001'\'' and time_index >=
'\''<previous_time>'\'' and time_index < '\''<current_time>'\''"}'
```

  - *<previous_time>* y *<current_time>* hay que sustituirlos por una fecha en formato ISO 8601, p. ej. 2024-04-04T17:34:26.000Z
  - **NO** hay cláusula GROUP BY
  - Resp.:

```
{ "cols": ["max"],
  "rows": [[2015]],
  "rowcount": 1,
  "duration": 271.6023 }
```

Time Series Data
Grafana

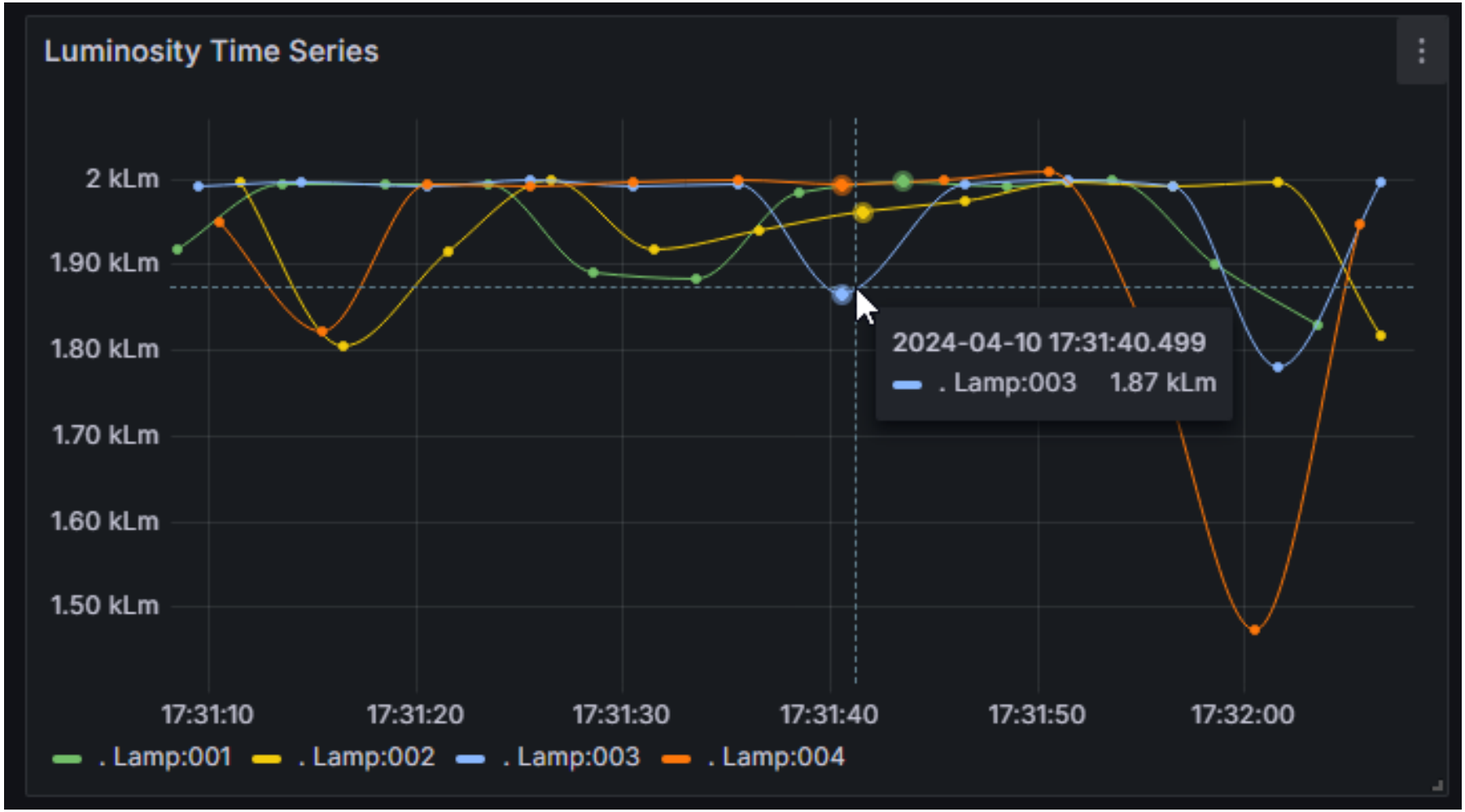
- **Conexión**
  - http://locahost:3003/
  - User: *admin*. Password: *admin*
- **Fuente de datos**
  - Menú izquierda → *Connections* → *Data sources*
  - *Add data source* → PostgreSQL
  - Parámetros
    - Name: CrateDB
    - Host: *crate-db:5432*
    - Database: *mtopeniot*
    - User: *crate*
    - TLS/SSL Mode: *Disable*
  - *Save & Test*

Time Series Data
Grafana

- **Crear *Dashboard***
  - Menú izquierda → *Dashboards* → *Create Dashboard*
  - Menú superior → *Save* → *Name:* "FIWARE Smart Store"
- **Crear un panel dentro del *dashboard: Add visualization***
  - Pestaña *Query*
    - *Data source*: CrateDB
    - *Format*: Time Series
    - *Table*: etlamp
    - *Columns*
      - time_index *Alias*: time
      - luminosity
      - entity_id
    - *Filter:* time_index*, Macros*: $__timeFilter
    - *Order*: time_index, ASC
    - NO *Limit*
  - *Panel type:* Time Series
  - *Panel options*
    - *Title*: Luminosity Time Series
    - *Line interpolation*: smooth
    - *Connect null values*: Threshold < 1h
    - *Unit*: Energy: Lumens

Time Series Data
Grafana

- **Panel *Luminosity Time Series***

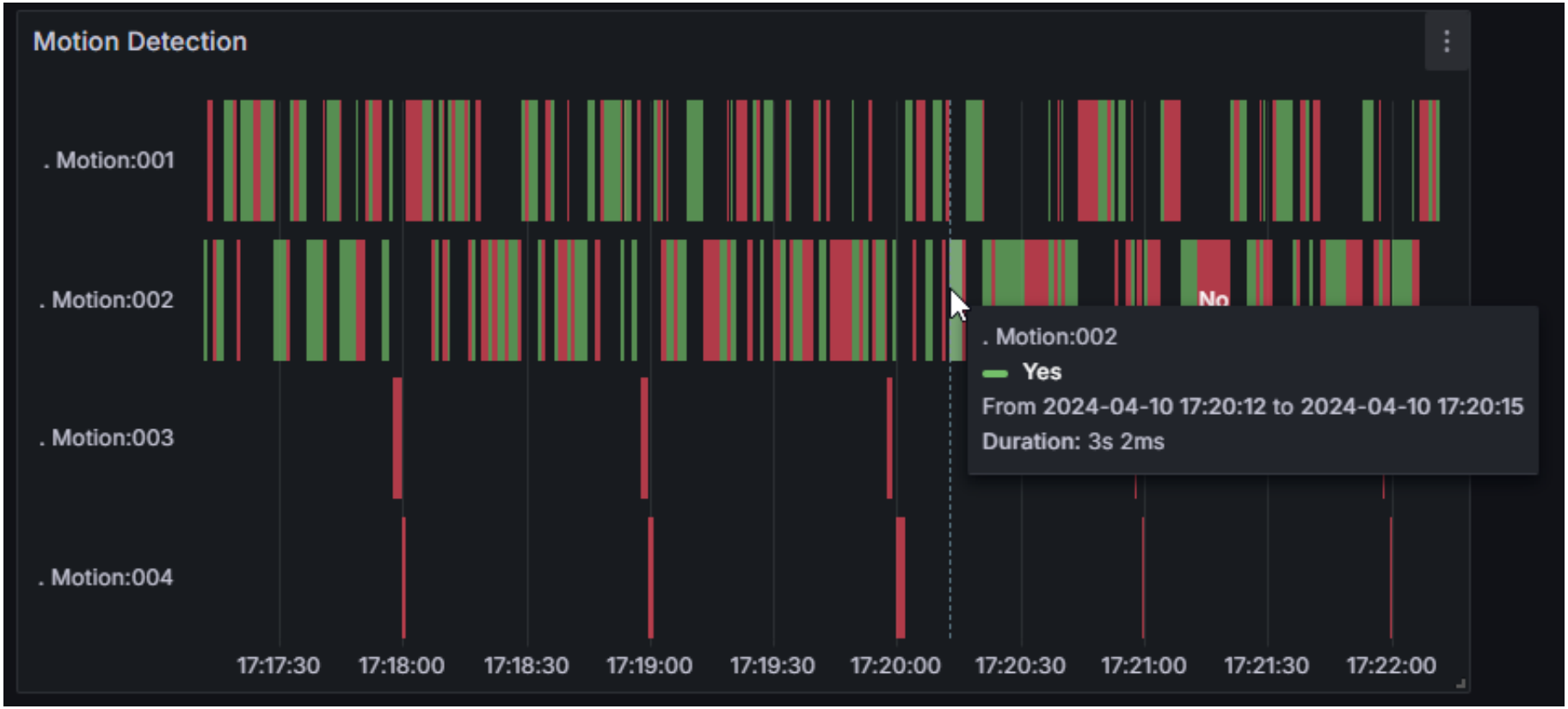


Time Series Data
Grafana

- **Panel *Motion Detection***
  - Menú superior → *Add visualization*
  - Tipo de visualización: *State timeline*
  - Pestaña *Query*
    - *Data source*: CrateDB
    - *Table*: etmotion
    - *Columns*:
      - time_index Alias: time
      - Count Alias: "."
      - entity_id
    - *Filter: time_index Macros*: $__timeFilter
    - *Format as*: Time series
  - *Panel options*
    - *Title*: Motion Detection
    - *Connect null values*: Always
    - *Align values*: Center
    - *Unit*: Boolean: On/Off
    - *Color scheme*: Red-Yellow-Green

Time Series Data
Grafana

- **Panel *Motion Detection***

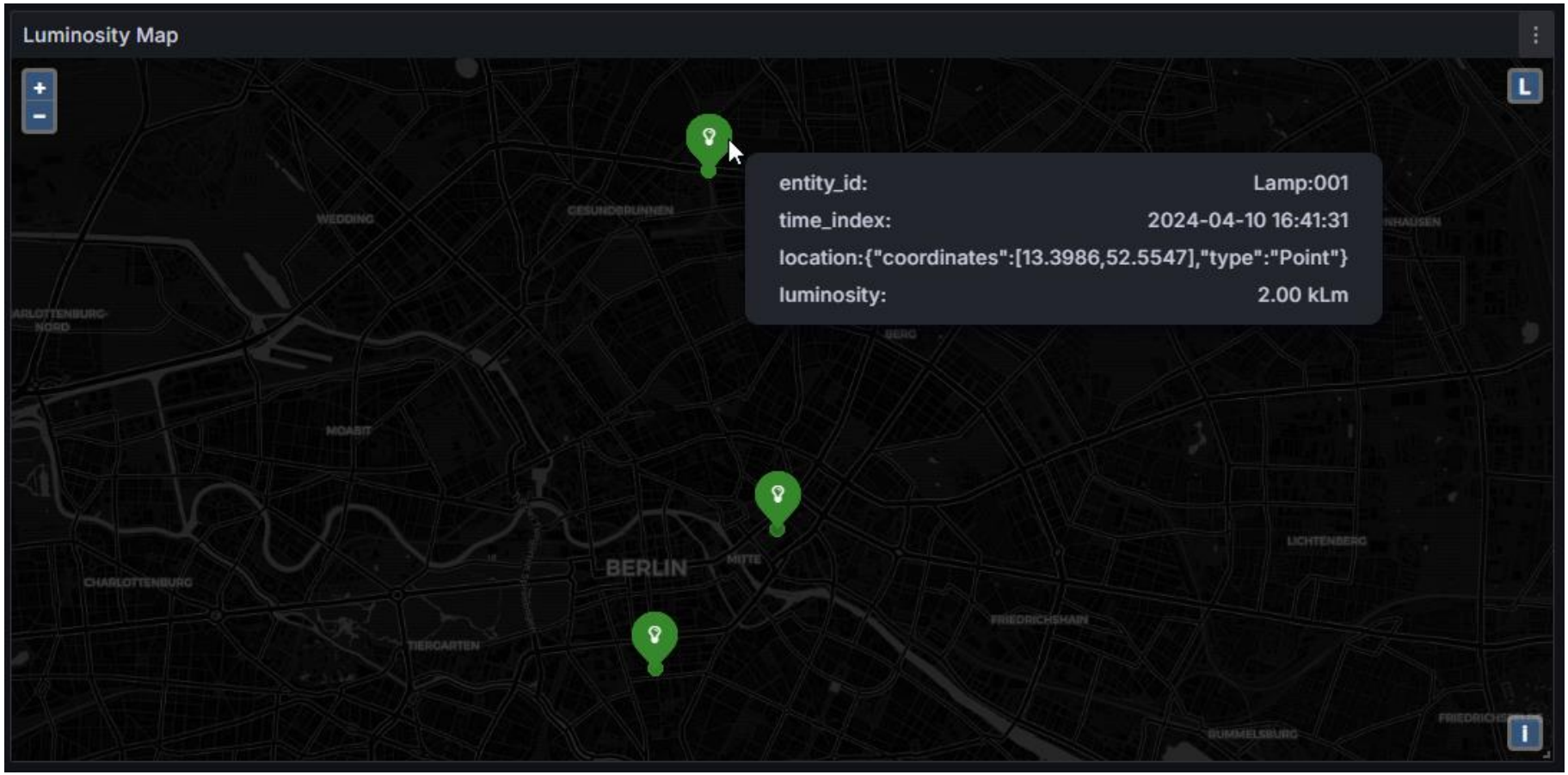


Time Series Data
Grafana

- **Panel *Luminosity Map***
  - Menú superior → *Add visualization*
  - Tipo de visualización: *Orchestra Cities Map*
  - Pestaña *Query*
    - *Data source*: CrateDB
    - *Format*: Table
    - *Table*: etlamp
    - *Columns:*
      - time_index
      - entity_id
      - location
      - luminosity
    - *Filter:* time_index*, Macros*: $__timeFilter
    - *Order*: time_index ASC
  - *Panel options*
    - *Title*: Luminosity Map
    - *Unit*: Energy: Lumens
    - *Add Layer*
      - *Query: A*
      - *Location: Auto*
      - *Show Pin:*
        - *Icon: lightbulb-o*
        - Pin size: 20
    - Hacer zoom hasta localizar las lámparas y entonces ir a *Panel options* → *Map view: Use current map settings*

Time Series Data
Grafana

- **Panel *Luminosity Map***



Time Series Data
Grafana

- **Exportar una *Dashboard* (JSON)**
  - Menú superior → Botón *Share* → *Export* → *Export for sharing externally*
- **Importar una *Dashboard* (JSON)**
  - Menú izquierda → *Dashboards* → *New (Import)*
  - Si existe una *Dashboard* con el mismo nombre / uid, hay que cambiarlos
  - Indicar la *data source* esperada por la *Dashboard* (crearla previamente)